# ECONOMETRIC RESEARCH BRIEF: RURAL–URBAN INFLATION PASS-THROUGH IN INDIA

**Gokhale Institute of Politics and Economics (GIPE)** | MoSPI CPI (Base 2012=100) Monthly Time Series (2014–2025, *N* = 142) | Methodology: OLS with Newey-West HAC Robust Inference

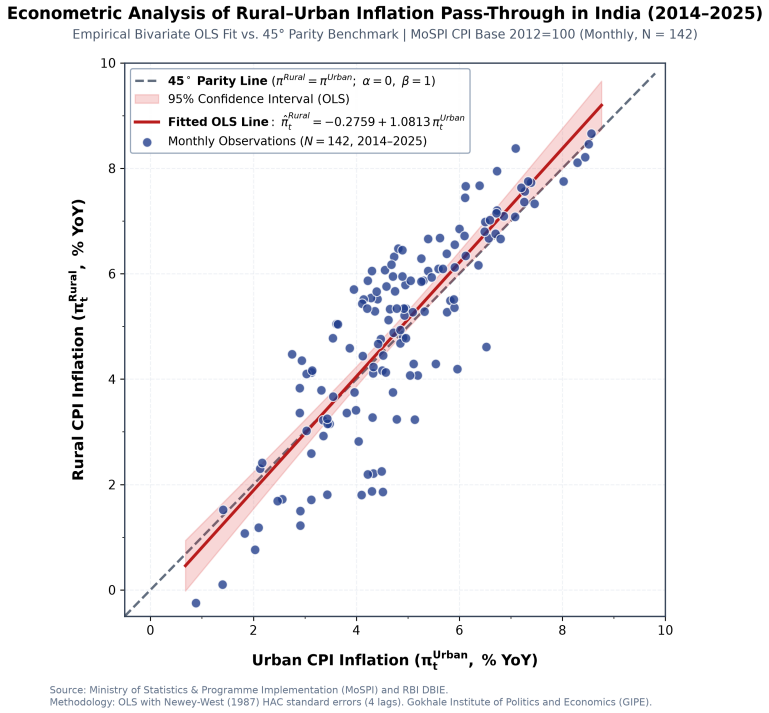



***Figure 1:*** *Publication Scatter Plot: Bivariate OLS Fit ($\pi\blacksquare^{Rural}$ = -0.2759 + 1.0813 $\pi^{Urban}$) vs. 45° Parity Line (β = 1, α = 0) with 95% Confidence Band.*

## Table 1: Econometric Model & Inference

| Regressor / Parameter | OLS Coef | OLS SE | HAC SE | HAC z | p-value |
|---|---|---|---|---|---|
| **Urban Inflation ($\beta_1$)** | **1.0813*** ** | 0.0549 | **0.0751** | 14.40 | **< 0.0001** |
| Intercept (α) | -0.2759 | 0.2776 | 0.4711 | -0.59 | 0.3219 |
| **$R^2$ = 0.7349** (Adj $R^2$ = 0.7330) | | σ = 1.00 35% | **DW = 0.1852** | | **N = 142** |

## Table 2: Consumption Basket Weights

| CPI Commodity Group | Rural | Urban | Wedge |
|---|---|---|---|
| **1. Food and Beverages** | **54.18%** | 36.29% | **+17.89 pp** |
| 2. Housing | 0.00% | 21.67% | -21.67 pp |
| 3. Fuel & Light | 7.94% | 5.58% | +2.36 pp |
| **Std Deviation (σ)** | **1.94%** | 1.54% | **+0.40 pp** |

**45° Parity Hypothesis Test (H■: α = 0, β = 1):** Joint Wald *F*(2, 140) = **2.0462** | ***p*-value = 0.1331** | **Decision: Fail to reject H■ at 5% significance level** (Critical *F* = 3.061). *Interpretation:* The empirical data shows long-run co-movement between rural and urban retail markets. The point estimate β■ = **1.0813** shows that rural inflation rises slightly faster during price increases.

**SUMMARY TAKEAWAYS:**

**1. Pass-Through Elasticity:** Bivariate OLS on monthly MoSPI CPI data (2014–2025, *N* = 142) yields an urban slope of β■ = **1.0813** (*p* < 0.0001; Newey-West HAC SE = 0.0751, *z* = 14.40), accounting for **73.5%** of variation in rural inflation.
**2. Parity Testing:** A joint Wald test fails to reject 45° parity [*F*(2, 140) = 2.046, *p* = 0.1331], confirming long-run spatial co-movement while capturing slight rural amplification (β > 1) during price increases.
**3. Basket Composition:** Food accounts for **54.18%** of rural CPI compared to **36.29%** in urban CPI (+17.89 pp), explaining higher rural volatility ($\sigma_{rural}$ = 1.94% vs. $\sigma_{urban}$ = 1.54%) and highlighting the role of supply-side agricultural policy alongside rate decisions.

Source: Ministry of Statistics & Programme Implementation (MoSPI) & RBI Database on Indian Economy (DBIE). Official GIPE Applied Econometrics Project.